College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Final Examination of Semester 1 Year: 2020

Subject: 353152 Electronic Circuits I Section: 5-6

Date: 2 November 2020 Time: 8.00-10.00

Name:______________________ ID:______________ Class:______________

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 10 pages (including this page) and they are worth 80 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

<u>Part 1</u> (50 points) Multiple choices

Instruction: Mark the correct answer for the following questions in **the provided answer sheet**.

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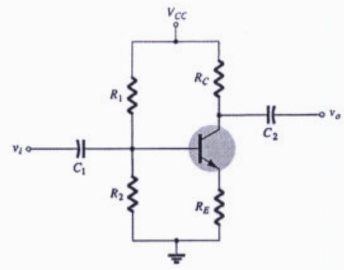
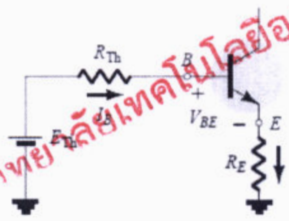
1. What is the DC-bias type of the circuit in Figure below?



(A) Fixed bias (B) Emitter-stabilized bias

(C) Voltage divider (D) Voltage feedback

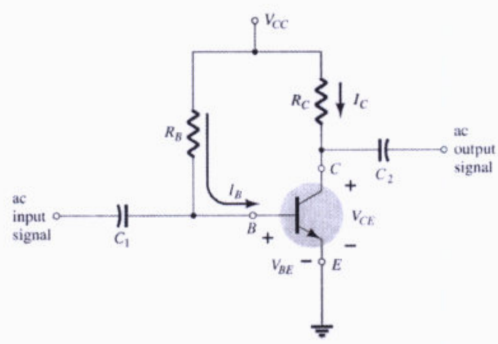
2. What is the KVL equation of the base-emitter loop of Figure below?



(A) $E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$ (B) $E_{Th} - I_B R_{Th} + V_{BE} - I_E R_E = 0$

(C) $E_{Th} + I_B R_{Th} + V_{BE} - I_E R_E = 0$ (D) $-E_{Th} - I_B R_{Th} - V_{BE} - I_E R_E = 0$

3. What is a bias type of the circuit in figure below.



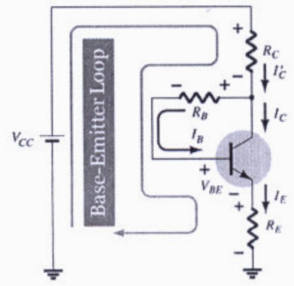
(A) Fixed-bias circuit (B) Emitter-stabilized bias circuit

(C) Voltage divider bias circuit (D) DC bias with voltage feedback

4. Which of the following transistor operating mode represents the short circuit?

(A) cut-off (B) saturation (C) active (D) open

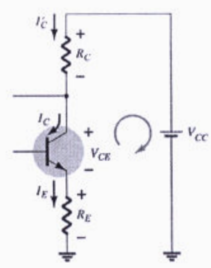
5. From Base-Emitter loop below, find the voltage equation of this closed loop.



(A) $I'_C R_C - I_B R_B - V_{BE} - I_E R_E = V_{CC}$ (B) $I'_C R_C - V_{BE} - I_E R_E = V_{CC}$

(C) $I'_C R_C + I_B R_B + V_{BE} + I_E R_E = V_{CC}$ (D) $I'_C R_C + V_{BE} + I_E R_E = V_{CC}$

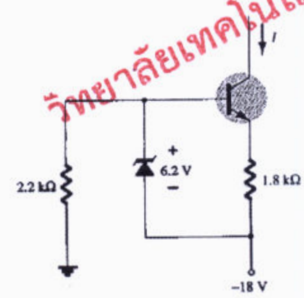
6. From Collector-Emitter loop below, find the voltage equation of this closed loop.



(A) $I_E R_E + V_{CE} + I_E R_C = V_{CC}$ (B) $I_E R_E - V_{CE} - I_E R_C = V_{CC}$

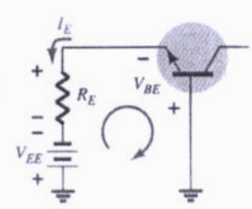
(C) $I_E R_E - V_{CE} + I_E R_C = V_{CC}$ (D) $I_E R_E - V_{CE} + I_E R_C = -V_{CC}$

7. Find constant current (A) of the transistor/Zener Constant-Current Source in Figure below.



(A) 3.06 mA (B) 1.06 mA (C) 2.06 mA (D) 5.06 mA

8. From base-Emitter loop below, find the KVL voltage equation of this closed loop.



(A) $V_{EE} + I_E R_E = V_{BE}$ (B) $V_{EE} - I_E R_E = -V_{BE}$

(C) $-V_{EE} + I_E R_E = V_{BE}$ (D) $V_{EE} - I_E R_E = V_{BE}$

9. In Collector-Feedback configuration, we assume that the $I'_C \cong I_C, I_E \cong I_C$. What is the reason of the assumption?

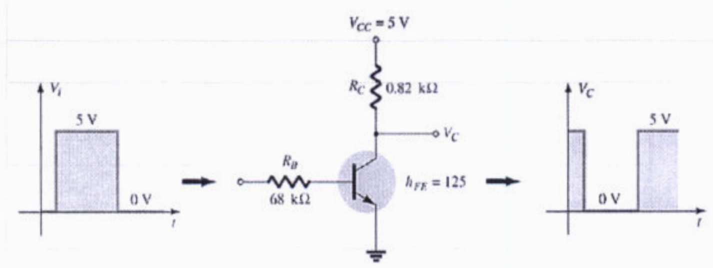
   (A) Because of $I_B << I_C$ (B) Because of $I_B = I_C$

   (C) Because of $I_B >> I_C$ (D) Because of $I_B \approx I_C$

10. In a fixed-bias circuit, the slope of the dc load line is controlled by _____.

    (A) R_B (B) R_C (C) V_{CC} (D) V_{BE}

11. From figure below, why output signal of this circuit is inverted.



    (A) 0V at input, the transistor is turn off and then Vout is Maximum.

    (B) 0V at input, the transistor is turn on and then Vout is Maximum.

    (C) 5V at input, the transistor is off and then Vout is Maximum.

    (D) 5V at input, the transistor is off and then Vout is Minimum.

12. Which class of power amplifiers is conducts less than 360° of input?

    (A) class A (B) class B (C) class AB (D) class C

13. Which class of power amplifiers does have Q-point at the middle of active region?

    (A) class A (B) class B (C) class AB (D) class C

14. How can the class A amplifier efficiency be improved from 25%?

    (A) increase input current. (B) reduce input current.

    (C) add transformer at input side. (D) add transformer at output side.

15. Which class of power amplifiers can have power efficiency up to 78.5%

    (A) class A (B) class B (C) class AB (D) class C

16. Which class of power amplifiers is used for the digital logic gate.

    (A) class A (B) class D (C) class AB (D) class C

17. Which type of following power amplifiers does have the highest efficiency?

    (A) class A (B) class B (C) class AB (D) class C

18. What is the propose of the transformer-coupled class A amplifier?

(A) Improve power efficiency (B) Reduce noise

(C) Reject undesired frequency (D) Increase stability factor

19. What is the value of $P_0(AC)$ of class B amplifier if $V_L(p\text{-}p) = 10$ V and $R_L = 2$ ohms.

(A) 50 W (B) 25 W (C) 6.25 W (D) 5.25 W

20. What is the purpose of using push-pull circuit for class B amplifier?

(A) to obtain full cycle (B) to reduce the distortion

(C) to increase speed of signal (D) to reduce the cost of amplifier

21. Which of the following is (are) the application(s) of a transistor?

(A) Signal protector (B) Voltage regulator

(C) Computer logic circuitry (D) DC Adapter

22. The _____the stability factor, the _____sensitive the network is to variations in that parameter.

(A) higher, more (B) higher, less (C) lower, more (D) None of the above

23. In a transistor-switching network, the level of the resistance between the collector and emitter is _____ at the saturation and is _____at the cutoff.

(A) low, low (B) low, high (C) high, high (D) high, low

24. Changes in temperature will affect the level of _____.

(A) current gain β (B) leakage current I_{CEO}

(C) both current gain β and leakage current I_{CEO} (D) None of the above

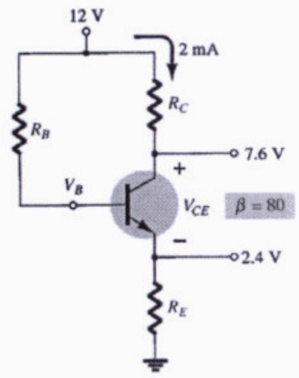
25. In a voltage-divider circuit, which one of the stability factors has the least effect on the device at very high temperature?

(A) $S(I_{CO})$ (B) $S(V_{BE})$ (C) $S(\beta)$ (D) Undefined

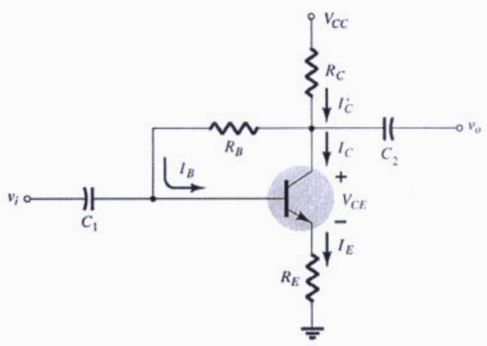
Part 2 (30 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

1. (5 points) From figure below, determine R_E.

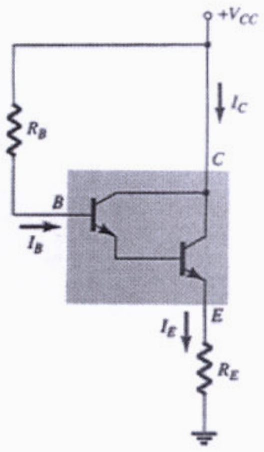


2. (5 points) Given R_C=6kΩ, R_B=150kΩ, R_E=5kΩ and V_{CC}=22V, find the current I_B and voltage V_{CE}.

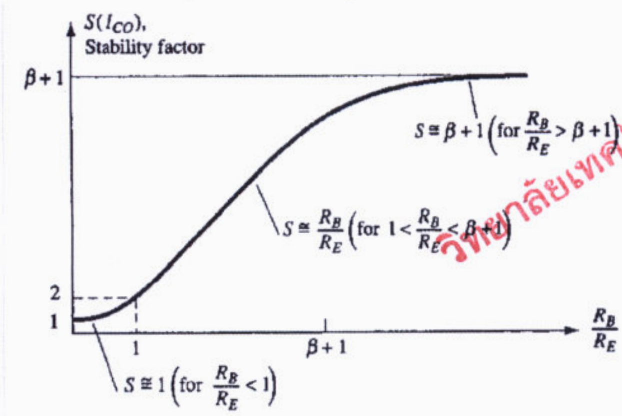


β=100

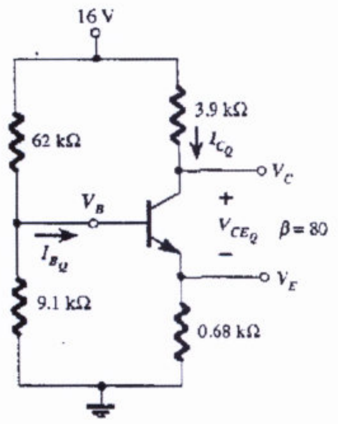
3. (5 points) Determine the dc bias voltages V_B and currents I_B and I_C in the circuit of Figure below. Given Vcc=18V, R_B=3.3 MΩ, β_D=8000,V_{BE}=1.6 V and R_E=390 Ω.



4. The stability factor $S(I_{CO})$ of Emitter-Bias configuration is shown in Figure below. Given R_B = 500 kΩ, R_E = 5 kΩ and β=100. What is the stability factor $S(I_{CO})$ value?



5. (5 points) Determine the I_{BQ}, I_{CQ} and $S(I_{CO})$ of the circuit of Figure below.



$$S(I_{CO}) = (\beta + 1)\frac{1 + R_{Th}/R_E}{(\beta + 1) + R_{Th}/R_E}$$

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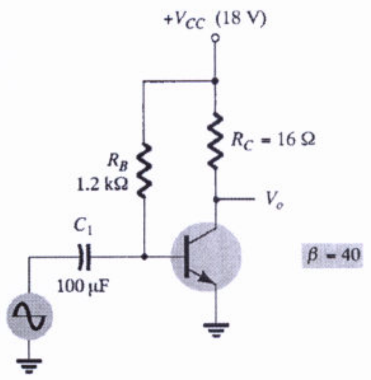
6. (5 points) Calculate the input and output power for the circuit below, if the input signal results in a base current of 4 mA(rms).



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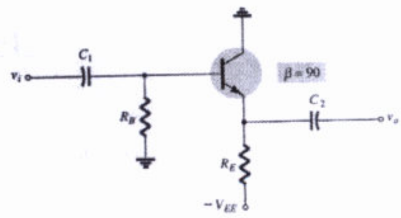
and Asst. Prof. Dr. Kittisak Phaebua

Handout

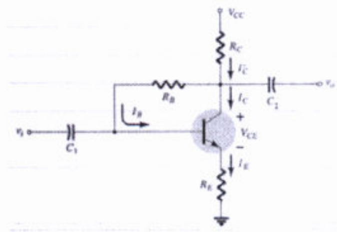
Emitter-follower



$$V_{CE} = V_{EE} - I_E R_E$$

$$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$$

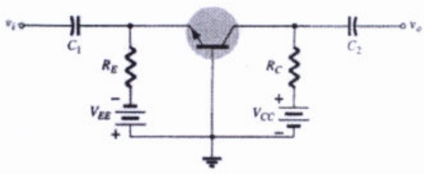
Collector-feedback



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Common-base



$$I_E = \frac{V_{EE} - V_{BE}}{R_E}$$

$$V_{CE} = V_{EE} + V_{CC} - I_E(R_C + R_E)$$

Class A amplifier

$$P_i(DC) = \frac{V_{CC}^2}{2R_C}$$

$$P_0(AC) = \frac{V_{CC}^2}{8R_C}$$

Transformer-couple class A amplifier

$$\%\eta = 50\left(\frac{V_{CE(\max)} - V_{CE(min)}}{V_{CE(max)} + V_{CE(\min)}}\right)^2$$

$$\frac{R_L}{R_L} = \frac{R_1}{R_2} = \left(\frac{N_1}{N_2}\right)^2 = a^2$$

Class B amplifier

$$P_0(AC) = \frac{V_L^2(p-p)}{8R_L} = \frac{V_L^2(p)}{2R_L}$$

$$P_i(DC) = V_{CC} I_{dc} = V_{CC}\left(\frac{2}{\pi} I_L(p)\right)$$

$$\%\eta = \frac{P_0(ac)}{P_i(dc)} \times 100\%$$

Answer Sheet

Name:______________________________ ID:________________

Subject: 342152 Electronic Circuits I

	(A)	(B)	(C)	(D)
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